

DUSSELDORF, Germany

WMO: 104000

Lat: 51.2958N

Lon: 6.7686E

Elev: 133

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.8	24.6	11.9	10.1	24.4	17.2	13.1	27.5	28.8	48.8	25.5	48.4	7.4	50	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	88.0	67.5	83.9	66.0	80.4	64.8	69.7	82.0	68.1	79.3	66.6	77.0	9.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	96.5	73.7	64.4	90.9	71.7	62.7	85.7	70.5	33.7	83.1	32.3	79.8	31.2	77.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.0	20.3	18.2	14.8	95.5	6.2	2.6	10.3	97.4	6.7	98.9	3.2	100.3	-1.3	102.2		
			13.8	73.1	6.0	1.9	9.5	74.4	6.0	75.5	2.6	76.5	-1.8	77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	37.8	38.7	44.3	51.9	57.6	63.9	67.8	65.6	60.6	53.8	45.6	40.6
	DBStd	12.33	7.54	7.20	7.33	6.87	7.00	6.05	6.32	5.50	5.54	6.67	6.59	7.21
	HDD50	1469	380	317	202	57	14	0	0	0	0	38	163	298
	HDD65	4897	843	736	641	394	247	92	42	56	157	351	581	757
	CDD50	2359	3	1	25	116	249	417	552	485	318	155	32	6
	CDD65	310	0	0	0	2	17	59	129	75	25	3	0	0
	CDH74	2754	0	0	0	58	177	552	1166	622	166	13	0	0
	CDH80	835	0	0	0	4	36	156	409	196	34	0	0	0
Wind	WSAvg	9.0	10.5	10.2	9.6	8.1	8.2	7.8	8.0	7.8	7.8	9.0	10.0	10.9
Precipitation	PrecAvg	34.0	3.0	2.6	2.3	2.1	2.5	3.1	3.3	3.5	2.6	2.7	3.1	3.3
	PrecMax	43.8	6.4	5.6	4.4	4.4	5.4	7.1	5.8	7.5	6.5	7.9	5.2	6.9
	PrecMin	27.0	0.2	0.4	0.4	0.0	0.7	0.8	1.2	1.2	0.7	1.1	0.2	1.1
	PrecStd	3.8	1.4	1.3	1.2	1.0	1.2	1.4	1.4	1.8	1.5	1.3	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	59.2	68.1	80.4	84.6	89.9	95.1	91.6	85.6	75.4	64.0	57.5
		MCWB	50.9	46.6	53.0	58.8	65.7	68.7	68.9	68.9	66.3	63.0	57.8	52.6
	2%	DB	53.6	53.4	63.8	75.0	78.7	84.3	89.5	85.7	78.8	69.9	60.5	55.3
		MCWB	49.8	47.2	51.1	57.5	61.9	66.1	67.6	68.0	64.8	60.6	55.2	51.4
	5%	DB	50.3	50.4	59.1	69.8	75.0	80.4	84.3	80.5	74.8	66.2	57.4	52.2
		MCWB	47.5	46.4	50.2	55.8	60.7	64.8	65.9	65.4	63.6	58.9	53.3	48.6
	10%	DB	48.2	48.4	55.5	64.7	71.3	76.6	80.5	76.7	70.1	62.8	55.4	50.0
		MCWB	45.7	45.1	48.4	53.5	58.7	63.1	65.4	63.9	61.4	57.5	51.5	47.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	50.9	55.4	60.9	67.9	71.3	72.2	72.5	68.8	63.9	58.2	54.1
		MCDB	55.4	54.3	62.7	74.0	80.4	84.3	86.5	84.9	79.9	72.3	62.5	56.9
	2%	WB	50.2	48.6	53.1	58.5	64.9	68.6	69.6	69.2	66.3	61.4	55.5	51.8
		MCDB	52.6	51.5	60.7	71.1	74.5	79.4	82.1	81.3	76.2	68.1	59.3	54.9
	5%	WB	47.9	47.0	51.0	56.5	62.1	66.5	68.2	67.0	64.3	59.7	53.9	49.7
		MCDB	50.5	50.4	57.4	67.7	71.0	76.5	79.6	77.3	72.1	65.1	57.5	52.2
	10%	WB	45.4	44.8	49.2	54.7	59.7	64.7	66.8	65.3	62.4	58.0	51.8	47.4
		MCDB	47.5	47.8	54.8	64.0	69.0	73.9	77.8	73.9	69.3	62.7	55.3	49.6
Mean Daily Temperature Range	5% DB	MDBR	8.8	10.0	14.4	18.1	17.5	17.6	17.6	16.7	15.8	12.8	9.5	8.0
		MCDBR	9.5	12.6	20.9	26.6	23.7	25.1	24.6	24.0	21.1	16.4	10.5	9.0
		MCWBR	8.5	8.6	11.9	12.8	11.2	11.0	8.4	10.1	10.0	9.3	7.8	7.3
	5% WB	MCDBR	9.3	11.3	17.5	23.1	20.0	20.6	20.7	19.8	18.4	14.2	9.2	8.6
		MCWBR	8.6	8.5	10.7	11.7	10.8	10.1	8.2	9.6	9.6	8.9	7.5	7.6
Clear-Sky Solar Irradiance	taub	0.326	0.349	0.378	0.414	0.419	0.419	0.428	0.420	0.392	0.364	0.342	0.321	
	taud	2.409	2.368	2.295	2.225	2.248	2.263	2.242	2.290	2.356	2.425	2.410	2.383	
	Ebn at Noon	219	244	258	261	266	266	261	257	251	236	209	199	
	Edh at Noon	19	26	34	40	41	41	41	37	31	24	19	17	
All-Sky Solar Radiation	RadAvg	233	447	802	1290	1554	1678	1596	1360	996	586	275	178	
	RadStd	36	85	122	147	142	150	199	127	119	55	36	24	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page